Digital Image Processing

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Submission deadline : 04/14 | Image Analysis & Image Data Compression | HomeWork 6

- Students are permitted a maximum of **8 day delays** throughout the course and you are allowed to use up to 5 days of your allowed delay per homework.
- Once the allowable delay period has expired, a **20% deduction** from the total assignment score will be applied **for each additional day of delay**.
- If external resources, such as **AI tools, websites or other references**, are utilized, proper **citation and attribution** are mandatory.
- Any instance of **plagiarism or cheating** will result in a score of **0** for all parties involved. A second violation will be considered **academic dishonesty** for the entire course, with further disciplinary actions as per institutional policies.
- Please upload all your answers in a file named **HW#_[SID]_[FullName].zip** to the **CW** system.

Theoretical (100)

1. Apply the following morphological operations to the given image and analyze the results.

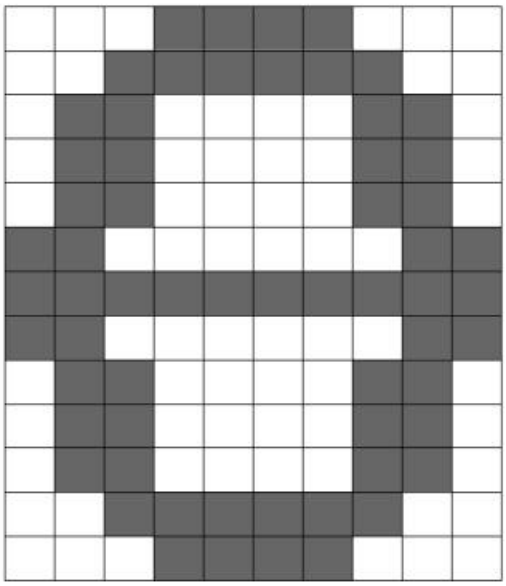
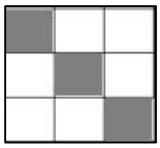
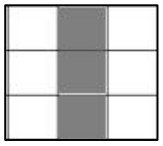


Figure 1: Binary representation of θ

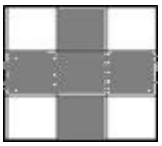
a) Apply dilation with below structuring element to the given image. (10)



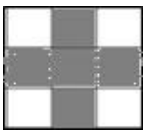
b) Apply erosion with below structuring element to the given image. (10)



c) Apply opening with a below structuring element to the given image. (10)



d) Apply closing with a below square structuring element to the given image. (10)



2. The following binary image represents the greek letter θ. (20)

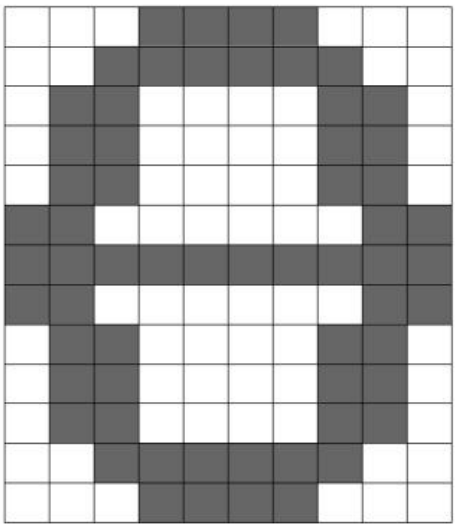


Figure 2: Binary representation of θ

What morphological operation(s) would you use to obtain the number zero (0) from this letter θ? Consider that you can use any of the following four structuring elements as many times as you wish (the "x" shows the central pixel).

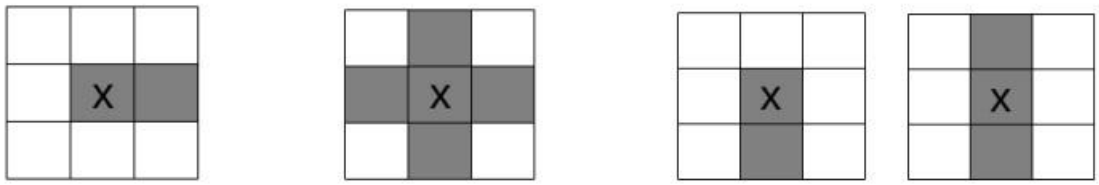


Figure 3: Structuring elements

Provide an explanation for your choice of morphological operation(s) as well as structuring element. You must show the output(s) of the morphological operation(s) that you perform in the following image(s).

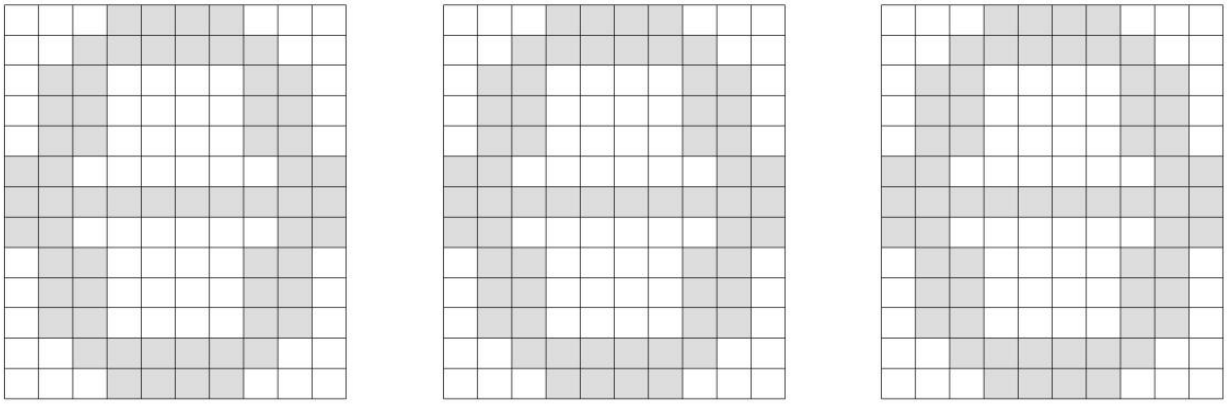


Figure 4: Images to be completed

3. Briefly describe each step of a transform coding system. Please reference the slides in your answer.(15)
4. Consider the given image:

0	0	2	2	7	7	7	7
0	0	2	0	3	6	6	4
0	2	1	3	6	6	5	4
1	1	1	6	5	5	4	4

   a) Compute its entropy. (5)
   b) Construct the Huffman code for this image, and determine the coding efficiency and the coding redundancy of this coding(5)
   c) Evaluate the efficiency of Huffman coding for this image.What should the characteristics of the image histogram be in order to be Huffman-coded efficiently?(5)
   d) What lossless technique could you use to make the image be more suitable for Huffman coding? Compute the resulting code.(5)
   e) How else would you compress this image without loss of information? Provide the compressed result. Identify the type(s) of redundancies targeted by your method.(5)